ANSWER KEY & MARKING SCHEME SET 1

Chapter 15: **The Making of a Scientist (Footprints Without Feet)** | Class 10 English | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** it linked Richard to a real research project	Connected him with an expert.
2	**(d)** early encouragement	Encouragement.
3	**(c)** diligent	Diligent.
4	**(d)** From Collector to Scientist	Reflects theme.
5	**(b)** questions arising from his observations	He asked 'why'.
6	**(b)** Because he read the book, Richard began tagging butterflies.	Because ... shows cause.
7	**(d)** to start something on one's own	Initiative.
8	**(b)** they respected his interests and gave resources	Supportive.
9	**(a)** should	Duty.
10	**(b)** greater confidence to pursue science	Confidence.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	To show that scientific interest begins early (1) and that collecting taught observation and classification (1).	1+1
12	Richard's parents supplied books and contacts (1) so his interest grew into a career (1).	1+1
13	He kept experimenting through many trials (1) and finally obtained significant results (1).	1+1
14	It gave him a real research task (1) and showed that a young person's data can matter to science (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	The chapter begins with childhood collecting and shows tagging monarchs. Then it describes school-level experiments and later research at Harvard. Each stage builds on the previous one.	1+1+1
16	He includes specific details like the book, the tagging and the projects. The story is told in simple chronological order. It builds admiration.	1+1+1
17	(a) What did Richard collect? (b) Who wrote to Dr Urquhart? / To whom did his mother write?	1+1+1
	OR: Richard's mother noticed his interest, gave him a book and wrote to an expert. His parents also allowed him time and space for projects. Their support was vital.	1+1+1
18	Students should be curious and ask questions. They should experiment and not accept ideas blindly. They should work with patience.	1+1+1
	OR: Richard loved butterflies, so he collected many of them. Both verbs are in the simple past.	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Define the statement (1). Early interests and collecting (1). Support from mother and mentor (1). Experiments and prizes (1). Conclusion (1).	2+2+1
	OR: Curiosity sets questions (1). Discipline carries out experiments (1). Examples (1). Results (1). Conclusion (1).	2+2+1
20	Question (1). Hypothesis (1). Experiment (1). Observation and result (1). Conclusion (1).	2+2+1
	OR: Greeting (1). Importance of science (1). Richard's example (1). Role of school and home (1). Conclusion (1).	2+2+1

Mistakes to avoid:

- Mixing the scientist's name: the chapter is on **Richard H. Ebright**, an American biologist, not on an Indian scientist.
- Saying he became a scientist only because of school; his **mother's support** (trips, telescopes, microscopes) and his collections drove his early curiosity.
- Forgetting the collections: from kindergarten he collected **butterflies**, then rocks, fossils and coins, and became an eager astronomer.

- Missing the turning point: a **book** on butterflies (and a later project on monarch butterflies) shaped his career; the chapter asks how a book changed his life.
- Calling him a famous scientist as a child; he was a shy boy who could not play team sports and so *collected things*.
- Spelling slips: *Ebright, scientist, curiosity, butterfly, telescope*; and tense mixing.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 15: **The Making of a Scientist (Footprints Without Feet)** | Class 10 English | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** an experiment on mimicry	Mimicry experiment.
2	**(a)** Richard was the best student of his time.	Judgement.
3	**(b)** his own experiments	Observation and experiment.
4	**(a)** misleading since they may have a function	Not mere decoration.
5	**(b)** advanced learning	A university.
6	**(a)** observe	Observe.
7	**(d)** gather data about movement	Data.
8	**(d)** tried every means to find answers	Thorough.
9	**(a)** mentor and learner	Guidance.
10	**(d)** active	He learned by doing.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	It supplies the facts that lead to questions (1) and helps check ideas against reality (1).	1+1
12	Others can verify and build on it (1) and shared data advances knowledge faster (1).	1+1
13	He asked a question and made a hypothesis (1) and tested it by experiment with careful observation (1).	1+1
14	They allowed him to present his results (1) and prizes boosted his confidence (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Science starts with questions, observation and experiment. It tests assumptions, such as the use of the gold spots. Patient work leads to discovery.	1+1+1
16	His hobby of butterfly collecting grew into serious research. As he gained skill, the hobby became his profession. The change was gradual.	1+1+1
17	Experiments test ideas against facts. Richard's experiment on the pupa showed a real function of the gold spots. Without it, the assumption would remain.	1+1+1
	OR: They develop skills like observation and patience. They can lead to a career. Richard's hobby of collecting led him to research.	1+1+1
18	Tagging lets scientists follow animals without harming them. It gives long-term data on movement. It is a simple and effective method.	1+1+1
	OR: He might not have met Dr Urquhart or joined the tagging. His curiosity might have faded. This shows the importance of encouragement.	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Introduction (1). Richard's hobby (1). Skills gained (1). Other examples (1). Conclusion (1).	2+2+1
	OR: Richard's activities (1). Your own hobbies (1). Similarities (1). Differences (1). What you can learn (1).	2+2+1
20	Mother's role (1). Dr Urquhart's role (1). Resources (1). Encouragement (1). Conclusion (1).	2+2+1
	OR: Prizes encourage (1). Richard's prizes (1). Real motive is curiosity (1). Risk of competition (1). Balanced conclusion (1).	2+2+1

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 15: **The Making of a Scientist (Footprints Without Feet)** | Class 10 English | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** pupa - stage between larva and adult	Metamorphosis.
2	**(c)** Richard's work was admired by many scientists.	Correct passive.
3	**(d)** one species resembling another for protection	Mimicry.
4	**(a)** Richard was given a book by his mother.	Passive.
5	**(c)** extraordinary	Extraordinary.
6	**(d)** recognition for original work	Recognition.
7	**(d)** Tagging butterflies was his hobby.	Tagging is a gerund.
8	**(c)** Do you think Richard's mother's role was crucial?	Judgement.
9	**(d)** collecting and classifying specimens	Collection.
10	**(c)** admiring	It admires him.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	It shows that curiosity and effort, not gender or wealth, make a scientist (1) and that support is available through parents and teachers (1).	1+1
12	People assumed the spots were only decoration (1) but testing proved they had a vital function (1).	1+1
13	Collecting gathers specimens (1) while research asks questions and tests answers (1).	1+1
14	He followed his interests with discipline (1) and used science to make real discoveries (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	The viceroy looks like the monarch which birds avoid. Richard tested whether this resemblance protects the viceroy. The experiment illustrates how species survive by mimicry.	1+1+1
16	(a) The butterflies were tagged by Richard. (b) He was awarded a prize by the fair.	1+1+1
17	It does not show failures directly but suggests long, patient work. Experiments need repetition. Persistence leads to results.	1+1+1
	OR: Dr Urquhart gave guidance on monarch tagging. His mother connected Richard with him. Mentors give direction and knowledge.	1+1+1
18	Dr Urquhart was a scientist who studied monarchs.	1+1+1
	OR: The word 'making' stresses process. A scientist is shaped by interest, encouragement and effort. The title fits the story of Richard.	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Collections (1). The book (1). Tagging (1). Experiments and fairs (1). Harvard (1).	2+2+1
	OR: Format (1). Reporter's questions (1). Richard's answers about tagging (1). Role of mother (1). Advice to students (1).	2+2+1
20	Address, date (1). Salutation (1). Story of Richard (1). Advice (1). Closing (1).	2+2+1
	OR: Opening (1). A key moment from the chapter (1). Personal link (1). Resolution to act (1). Concluding line (1).	2+2+1

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